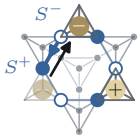


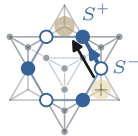
i.a) create


 $S_i^+ S_j^-$: pair splits

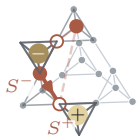
i.b) move


 $S_i^+ S_j^-$: spinon walks

i.c) annihilate


 $S_i^+ S_j^-$: + meets -
 $\mathbf{q}_\gamma = (0, 0, 0)$

i.d) create

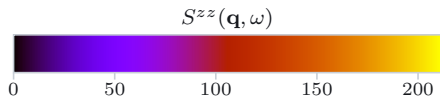

 $S_i^+ S_j^-$: pair splits

i.e) separate

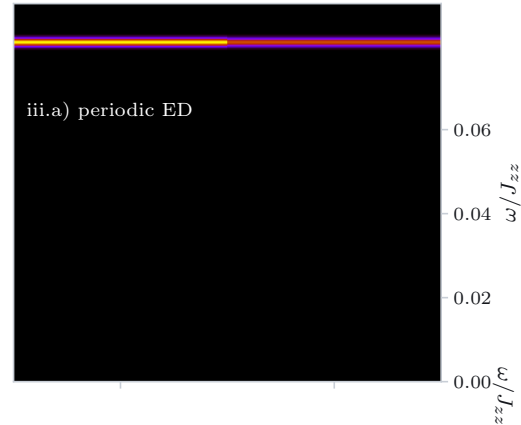


+ carried one period down

i.f) annihilate


the boundary closes it
 $\mathbf{q}_\gamma = (0, 0, -1)$


iii.a) periodic ED



iii.b) winding-free

